

8 (**short**) As shown in Fig. 5, a rigid uniform bar of mass m and length l hangs vertically under gravity from a frictionless hinge located at the top end of the bar. The bar is initially at rest. The bar is then impacted by a projectile moving horizontally at speed v and mass m at a distance $2l/3$ below the hinge. The projectile remains attached to the bar after impact.

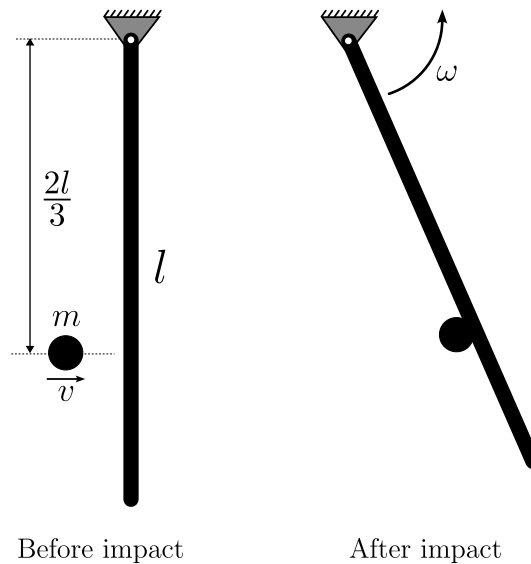


Fig. 5

- (a) Find the angular velocity ω of the bar just after impact. [5]
- (b) Find an expression for the speed v of the projectile that is required to rotate the bar at least 180 degrees after impact, i.e. vertically above the hinge. [5]